

Chapter 9

Natural Vegetation

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Which of the following is true about natural vegetation?

1. It refers to the plant community which has grown naturally.
2. It has not been disturbed over a long time.
3. It allows its individual species to adjust themselves to the climate and soil conditions.
4. All of the above.

Answer

All of the above

Question 2

Plants of a particular region or period, listed as species and considered as a group is called

1. Flora
2. Vegetation
3. Forest
4. All of the above.

Answer

Flora

Question 3

The assemblage of plant species living in association with each other in a given environmental set-up is called

1. Forest

2. Vegetation
3. Flora
4. None of the above.

Answer

Vegetation

Question 4

What are the chief characteristics of Tropical Rain forests?

1. They appear green all the year round.
2. They have dense canopy of trees.
3. The trees do not have a fixed time to shed their leaves.
4. All of the above.

Answer

All of the above.

Question 5

Why are the Tropical Evergreen forests not fully exploited?

1. Because they do not shed their leaves and have dense canopy of trees.
2. Because they are found in highly inaccessible areas.
3. Because of tangled mass of canes, palms and bamboos.
4. Because they have dense undergrowth.

Answer

Because of tangled mass of canes, palms and bamboos.

Question 6

Tropical Deciduous forests are also known as

1. Rain forests
2. Monsoon forests
3. Tidal forests
4. None of the above

Answer

Monsoon forests

Question 7

Sandalwood belongs to category of forests.

1. the Moist Deciduous forests
2. the Dry Deciduous forests
3. the Rain forests
4. the Mountain forests

Answer

the Moist Deciduous forests

Question 8

The type of vegetation found in Tropical Thorn forests is

1. Thyrophytic
2. Thornyphytic
3. Xerophytic
4. Phylophytic

Answer

Xerophytic

Question 9

At which of the following places Tidal forests are found?

1. Wet marshy areas
2. River deltas and along sea coasts
3. Swampy areas
4. All of the above

Answer

All of the above

Question 10

Why do Tidal forests have a profuse growth with tangle of climbers?

1. To adapt themselves to survive in soft and shifting mud in wet marshy areas and river deltas.
2. To breathe while being submerged under water through pores.
3. To adapt themselves to live in damp and moist conditions.
4. None of the above.

Answer

To adapt themselves to survive in soft and shifting mud in wet marshy areas and river deltas.

Question 11

Littoral forests have breathing roots called

1. Pneumatophores
2. Pinophores
3. Phytopores
4. None of the above.

Answer

Pneumatophores

Question 12

Mangrove trees are utilised for which purpose?

1. Timber
2. Fruit
3. Ornamental objects
4. Fuel

Answer

Fuel

Question 13

For which purpose are Sundari trees utilised?

1. Rearing shellac worms.
2. Furniture, railway carriages and bridges.
3. Hard durable timber for construction purposes and boat making.
4. None of the above.

Answer

Hard durable timber for construction purposes and boat making.

Question 14

Why do tropical deserts have stunted growth?

1. Due to high temperature these trees do not grow to great heights.
2. Due to scarcity of rain water the trees do not get enough water for their growth.
3. Due to long roots that reach deep underground and therefore do not attain height.
4. None of the above.

Answer

Due to scarcity of rain water the trees do not get enough water for their growth.

Answer the following questions

Question 1(a)

What is meant by natural vegetation?

Answer

Natural vegetation refers to the plant community which has grown naturally without any human assistance and has not been disturbed over a long time, so as to allow its individual species to adjust themselves fully to the climate and soil conditions.

Question 1(b)

Give two points of difference between vegetation and forests.

Answer

Vegetation	Forests
Vegetation refers to the assemblage of plant species living in association with each other in a given environmental set-up.	Forest refers to a large tract of land covered with accompanying undergrowth of shrubs, herbs and thousands of life forms.
Vegetation includes only plant species.	Forests sustain both plant and animals.

Question 1(c)

Give a geographical reason for each of the following:

- (i) The Western Ghats are covered with thick evergreen forests while the Eastern Ghats are covered by deciduous forests.
- (ii) Tropical evergreen forests have tall trees and appear green all the year round.
- (iii) Deciduous forests are commercially most exploited.

Answer

(i) Evergreen forests are found in areas of high temperature, having high annual rainfall. Western Ghats receive more than 200 cms. of annual rainfall with an average annual temperature between 25°C and 27°C. Similarly, deciduous forests are found in areas with moderate rainfall and the eastern ghats receive annual rainfall between 100 cm and 200 cm with mean annual temperature range of 24°C and 27°C. Hence, the Western Ghats are covered with thick evergreen forests while the Eastern Ghats are covered by deciduous forests.

(ii) Tropical evergreen forests have tall trees as this region remains warm and wet throughout the year. Since the trees do not have a fixed time to shed their leaves, to flower or for fruition, the forest appears green all the year round.

(iii) Deciduous forests are commercially most exploited because besides providing valuable timber, they provide various other valuable products, like sandalwood, which are in great demand in India and abroad. Moreover, they are less dense which makes transport between these forests and markets easier and economical.

Question 1(d)

- (i) Name one area where tropical rain forests are found.
- (ii) Mention two main characteristics of tropical rain forests.

Answer

(i) Tropical rain forests are found on the western slopes of the Western Ghats.

(ii) Two main characteristics of tropical rain forests are-

1. Trees reach great heights of more than 60 m.
2. Since this region remains warm and wet throughout the year, it has a luxuriant vegetation of all kinds- trees, shrubs and creepers which give it a multi-layered structure.

Question 2(a)

Give two points of difference between Tropical Evergreen and Tropical Deciduous Forests.

Answer

Tropical Evergreen Forests	Tropical Deciduous Forests
Tropical Evergreen forests are found in areas which receive more than 200 cms. of annual rainfall.	Tropical Deciduous Forests are found in areas which receive annual rainfall between 100 cm and 200 cm.
Trees in tropical evergreen forests do not shed their leaves.	Trees in deciduous forests shed their leaves for a few weeks during spring and early summer.

Question 2(b)

What are the climatic conditions under which Tropical Evergreen Forests develop?

Answer

The climatic conditions under which Tropical Evergreen Forests develop are-

1. Annual rainfall of more than 200 cm
2. Average annual temperature between 25°C and 27°C
3. Average annual humidity exceeds 77 percent

Question 2(c)

Give a geographical reason for each of the following:

- (i) Tropical Evergreen Forests are difficult to exploit for commercial purposes.
- (ii) Delta area of river Ganga is called Sunderbans.
- (iii) Forests are grown in and around the cities having Iron and Steel Industries, and Thermal Power Plants.

Answer

- (i) Tropical Evergreen Forests are difficult to exploit for commercial purposes because of the tangled mass of canes, palms, bamboos, ferns and climbers along with the lack of means of transport.
- (ii) Delta area of river Ganga is called Sunderbans because the forests in this area have sundari trees.

(iii) Forests are grown in and around the cities having Iron and Steel Industries, and Thermal Power Plants because the level of pollution is much higher in areas where industries are located and forests prevent pollution and act as a barrier for the dust and purify the air.

Question 2(d)

(i) What are the characteristic features of deciduous forests?

(ii) Name the two types of deciduous forests.

Answer

(i) The characteristic features of deciduous forests are-

1. The trees in these forests shed their leaves from six to eight weeks during spring and early summer.
2. A particular species can be found over a large area.
3. They are commercially most exploited.

(ii) The two types of deciduous forests are-

1. Moist Deciduous Forests
2. Dry Deciduous Forests

Question 3(a)

Mention the areas where Tropical Desert Vegetation thrives.

Answer

Tropical Desert Vegetation thrives in south-western Punjab, Haryana, Uttar Pradesh, central and eastern Rajasthan, Madhya Pradesh, Chhattisgarh, Gujarat, parts of Maharashtra, Karnataka and Andhra Pradesh.

Question 3(b)

Under what climatic conditions do Littoral forests develop?

Answer

These forests thrive in areas where temperature remains between 26°C to 29°C and the rainfall is not more than 200 cm.

Question 3(c)

Give a geographical reason for each of the following:

- (i) The forest area has greatly depleted in India.
- (ii) The Tropical Evergreen forests are found on the western slope of the Western Ghats.
- (iii) Acacia has long roots.

Answer

- (i) The forest area has greatly depleted in India as forests have been cleared for the purpose of cultivation. Timber has become an important commercial resource. Rapid industrialization, urbanisation and expansion of cities has also led to the destruction of forest cover in the country.
- (ii) The Tropical Evergreen forests are found on the western slope of the Western Ghats because evergreen forests are found in areas of high temperature and high annual rainfall. Western Ghats receive more than 200 cms. of annual rainfall with an average annual temperature between 25°C and 27°C.
- (iii) Acacia has long roots so that the roots can reach deep, ground water resources and therefore, survive drought like conditions.

Question 3(d)

- (i) State two characteristic features of tropical desert vegetation. Name two typical trees found here.
- (ii) Why tropical deserts have stunted trees.

Answer

- (i) Two characteristic features of tropical desert vegetation are-
 1. These forests have Xerophytic vegetation. Due to paucity of rainfall, the trees are stunted with large patches of coarse grasses.
 2. Plants remain leafless for most part of the year and look like scrub vegetation.

Two typical trees found here are babool and date palm.

- (ii) Tropical deserts have stunted trees due to paucity of rainfall.

Question 4(a)

What are 'Tidal forests'? Name two typical trees found there.

Answer

Tidal forests are one of the most specialised tropical natural vegetation types. India has a variety of wetlands habitats. These lands are between terrestrial and aquatic system where the water table is usually at or near the surface of land.

Two typical trees found here are Keora and Amur.

Question 4(b)

Name the type of forests found in the western part of the Western Ghats. Give two reasons why these forests are so named.

Answer

Tropical Evergreen forests are found in the western part of the Western Ghats. They are named so because they remain green throughout the year as they do not have any particular season to shed their leaves and also because they grow in the regions receiving annual rainfall of more than 200 cm.

Question 4(c)

Give a geographical reason for each of the following:

- (i) Forests constitute an important natural resource.
- (ii) Agro forestry is not vulnerable to population stress.
- (iii) Tidal forests have profuse growth with tangle of climbers.

Answer

(i) Forests constitute an important natural resource because trees provide us with products such as fruits, leaves, roots and tubers of plants. Wood is used for making furniture and as a source of energy. Wood and bamboo pulp is used to make paper. Forests also provide fibers, essential oils, edible plants etc.

(ii) Agro forestry is not vulnerable to population stress as agro forestry co-ordinates with the local population and reduces the pressure of population on traditional forestry.

(iii) Tidal forests have profuse growth with tangle of climbers to survive in soft and shifting mud.

Question 4(d)

Describe the methods of protecting forests under the following heads:

- (i) Afforestation.
- (ii) Using alternative sources of energy.

Answer

(i) Afforestation means planting trees on lands which are not previously covered with forest vegetation.

(ii) To conserve forests, we can use non-conventional or renewable sources of energy like solar energy, tidal energy, hydel energy etc.

Question 5(a)

What is social forestry?

Answer

Social forestry refers to the management and protection of forests and afforestation on barren lands with the purpose of helping in the environmental, social and rural development.

Question 5(b)

Give two points of difference between agro-forestry and traditional forestry.

Answer

Agro-forestry	Traditional forestry
Agro forestry provides a combined produce of forest and agricultural crops.	Traditional forestry only provides the products.
It is not vulnerable to population stress.	It is vulnerable to population stress.

Question 5(c)

Mention any three methods for the conservation and development of forests in India.

Answer

Three methods for the conservation and development of forests in India are-

1. Increasing the area under forests
2. Afforestation around industrial units
3. Using alternative sources of energy

Question 5(d)

State three objectives of agro-forestry.

Answer

Three objectives of agro-forestry are-

1. To reduce pressure on natural forests for obtaining timber as well as non-timber forest produce.
2. To check soil erosion and to maintain the natural fertility of the soil.
3. To maintain ecological balance along with proper utilisation of farm resources.

Question 6(a)

Why does the vegetation in the Himalayas vary with altitude?

Answer

The vegetation in the Himalayas vary with altitude because in the mountain regions, temperature decreases with altitude.

Question 6(b)

Why does Tropical Evergreen forests appear dark inside the forests.

Answer

Tropical Evergreen forests appear dark inside the forests because of the dense canopy of trees which do not allow enough sunlight to reach the ground.

Question 6(c)

Give a geographical reason for each of the following:

- (i) Nilgiri region has variety of vegetation type.
- (ii) Sundari trees are used for boat making and construction purposes.
- (iii) Littoral forest is also called tidal forest.

Answer

(i) In mountain regions, temperature decreases with increase in altitude. This has a corresponding effect on the natural vegetation of the mountain region. Thus, vegetation in lower regions of Nilgiri is sub tropical and vegetation in upper regions of Nilgiri is temperate.

(ii) Sundari trees are used for boat making and construction purposes because these trees provide hard durable timber.

(iii) Littoral forest is also called tidal forest because they are found in tidal zone, an area that is above water at low tide and under water at high tide.

Question 6(d)

Explain how is man responsible for the reduction of forest cover in India.

Answer

The felling of the trees for the purpose of obtaining timber is an important cause of reduced forest cover in India. Rapid industrialization, urbanisation and expansion of cities has also led to the destruction of forest cover in the country.

Question 7(a)

Name two areas where Tropical deciduous vegetation is found.

Answer

Tropical deciduous vegetation is found in north-eastern part of the peninsula and along the foothills of the Himalayas.

Question 7(b)

Why do the Coniferous trees of mountain forests have conical shape?

Answer

Since coniferous trees are found in cold regions, they are well adapted to the cold climate. Their leaves are conical in shape so that the snow slides off easily from the branches without causing any damage to them.

Question 7(c)

Give a geographical reason for each of the following:

- (i) Littoral forest vegetation has breathing roots.
- (ii) Forests are often compared to human lungs.
- (iii) Van Mahotsav is celebrated in our country every year.

Answer

(i) Littoral forest vegetation has breathing roots because these root enable the plants to breathe during the waterlogged conditions which occur when high tide comes.

(ii) Forests are compared to human lungs as the trees which make up a forest act as the compartment of lungs which help in inhalation and exhalation of the air. But in lungs, Oxygen is taken in and Carbon dioxide is given out whereas in trees carbon dioxide is inhaled and oxygen is exhaled.

(iii) Van Mahotsav is celebrated in our country every year to create awareness about the importance of trees and encourage community participation in planting trees.

Question 7(d)

What is meant by 'farm forestry'? State two of its advantages for farmers.

Answer

Farm forestry is the practice of growing trees on the farm land to produce saleable products like timber, tannin, charcoal, etc. and to provide shade and shelter for stock and crops.

Two advantages of farm forestry are-

1. It increases farm income.
2. It supports community development.

Thinking Skills

Question 1

You went for a study tour to Mahabaleshwar in the Western Ghats. On reaching there, you found a luxuriant vegetation and evergreen trees. What accounts for such a growth of vegetation? What type of forests did you find in the area? Name any three types of trees found there and their economic importance.

Answer

Recently I went to Mahabaleshwar for a study tour and found luxuriant vegetation and evergreen trees. My teacher explained that such vegetation and evergreen trees are found in the areas where the annual rainfall is more than 200 cm with a short dry season, average annual temperature is between 25°C and 27°C and average annual humidity exceeds 77 percent.

I found tropical evergreen forests there. The main variety of trees found in these forests include rosewood, ebony and mahogany. Rosewood is used for making furniture, floor board wagon parts, etc. It is used as decorative wood for carving and for making ornamental ply-boards. Ebony is used for ornamental carving and for decoration. It is used for inlaying, for making musical instruments, sports goods, piano keys and caskets. Thus, the timber produced by evergreen trees is of high economic value.

Question 2

You live in an area which is surrounded by the commercially most exploited forests. Which forests are these and why are they most exploited forests in India? Besides your area, where are these forests found? State the climatic conditions required for such forests to flourish.

Answer

Tropical Moist Deciduous Forests are found in the area where I reside. These are commercially the most exploited forests because they provide invaluable timber and various other products. They are less dense which makes transport easier and economical. Sandalwood found in these forests is in great demand in India and abroad.

Besides my area, these forests occur in the north eastern part of the Peninsula, along the foothills of the Himalayas and eastern slopes of the Western Ghats. They occupy a sizeable area in Uttar Pradesh, Maharashtra, Karnataka and Tamil Nadu.

Tropical Moist Deciduous Forests are found in areas with moderate or low annual rainfall of 100 cm to 200 cm and the mean annual temperature of 24°C and 27°C and humidity percentage of 50 to 80.

Question 3

Imagine that you want to start a factory to manufacture plant based medicines, beauty products and toothpastes. Which forest area would you choose in the vicinity of which you can start your factory and why?

Answer

I would choose to start my factory in the vicinity of tropical desert forest because neem, babool and date palm trees are found in these forests.

Babool bark and gum and Neem bark and roots have medicinal properties. It is effective in treating diabetes, allergies, ulcers and several other diseases. Hence, it can be used for making plant-based medicines. Date palm is eaten raw and also used as an astringent, as a decoction syrup or paste for sore throat, cold, fever, etc. Neem oil, leaves and neem extracts are used to manufacture health and beauty products. Neem is also used as an insecticide. Thus, beauty products and toothpastes can also be manufactured by using these trees.

Question 4

If given a choice between agro-forestry and traditional forestry, which one will you choose? Why?

Answer

I would choose agro-forestry over traditional forestry because it aims to provide conservation of the land and its improvement in order to achieve a combined produce of forest and agricultural crops.

